

Isotopes and Average Atomic Mass

Answer key with worked solutions

A. Understanding

1. What the atomic mass represents.

The **weighted average** mass of the element's naturally occurring isotopes, based on their abundances.

2. Which isotope is more abundant?

The **20.0 u** isotope: the average (20.2 u) is much closer to 20.0 u than to 22.0 u, so the lighter isotope must be far more common.

B. Calculating average atomic mass

Each abundance is written as a fraction (percentage $\div$ 100), which has no unit.

3. Boron.

$$(0.199 \times 10.013 \text{ u}) + (0.801 \times 11.009 \text{ u}) = \mathbf{10.81 \text{ u}}$$

4. Rubidium.

$$(0.7217 \times 84.912 \text{ u}) + (0.2783 \times 86.909 \text{ u}) = \mathbf{85.47 \text{ u}}$$

5. Silver.

$$(0.5184 \times 106.905 \text{ u}) + (0.4816 \times 108.905 \text{ u}) = \mathbf{107.87 \text{ u}}$$

6. Silicon.

$$(0.9223 \times 27.977 \text{ u}) + (0.0468 \times 28.976 \text{ u}) + (0.0309 \times 29.974 \text{ u}) = \mathbf{28.09 \text{ u}}$$

C. Challenge

7. Abundances of copper isotopes.

Let x = fraction of copper-63 (no unit): $(62.930 \text{ u})x + (64.928 \text{ u})(1 - x) = 63.546 \text{ u}$

$64.928 \text{ u} - (1.998 \text{ u})x = 63.546 \text{ u}$, so $x = \frac{1.382 \text{ u}}{1.998 \text{ u}} = 0.692$

Copper-63: 69.2%, copper-65: 30.8%